ALEXANDER POLYNOMIAL

SEIFERT SURFACES

EXAMPLES

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DEFINITION

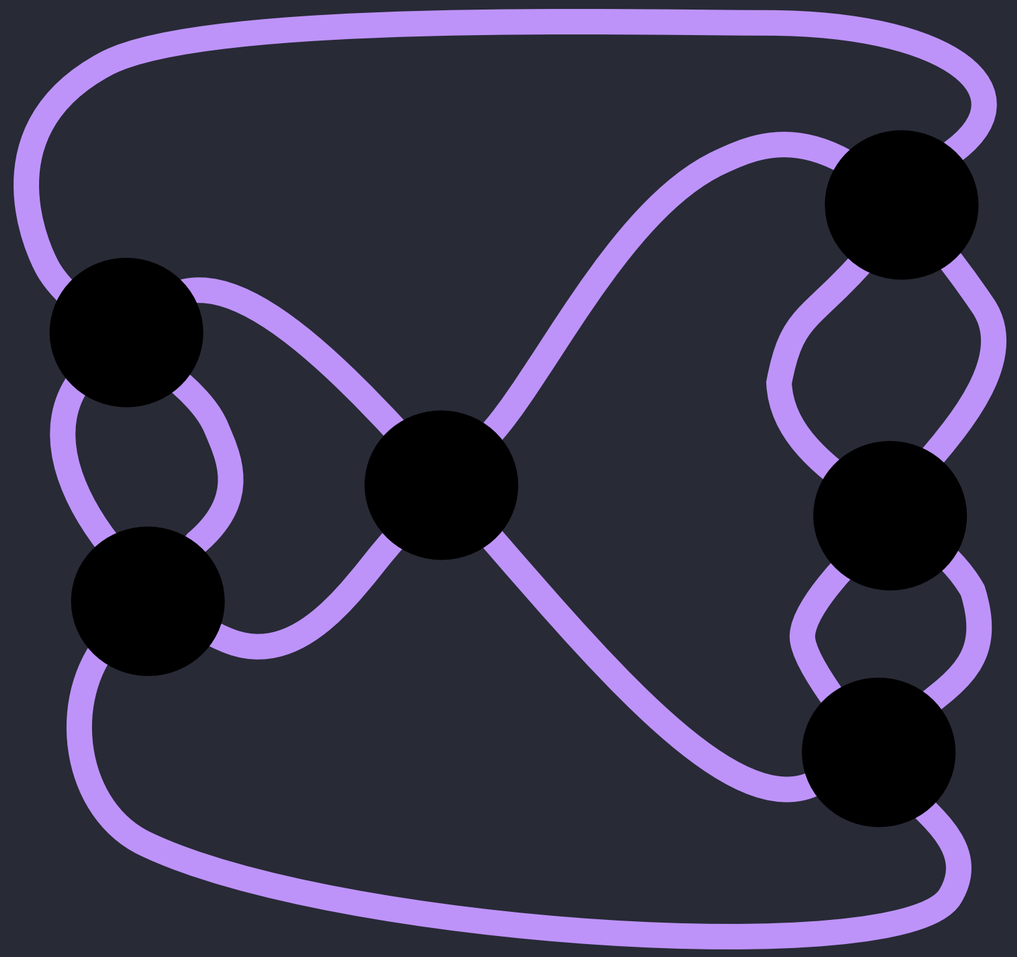
A Seifert surface for an oriented link in S^3 is a compact connected oriented surface smoothly embedded in S^3 with oriented boundary equal to the link.

EXISTENCE

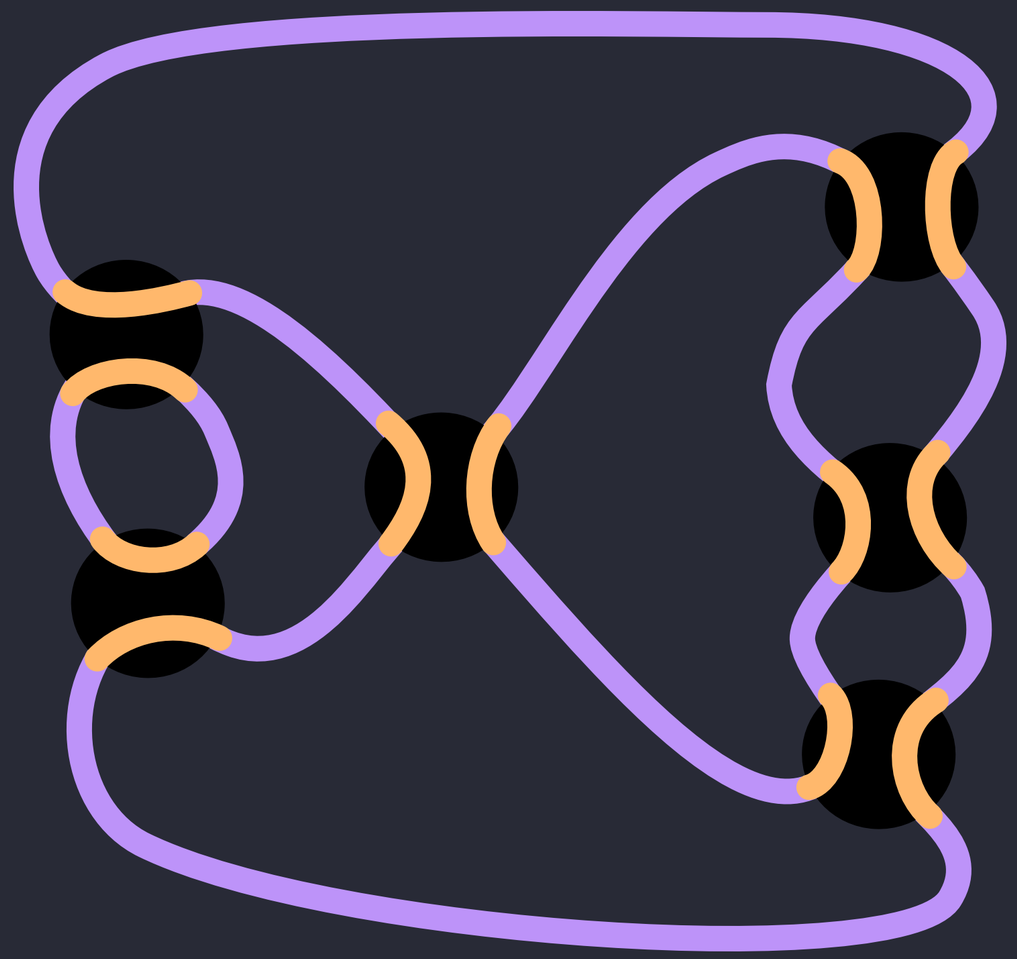
Existence can be shown by an algorithm to construct a Seifert surface from a given link projection.

REMOVE CROSSINGS

CLOSE CURVES



COLLECTION OF DISKS



ATTACHING BANDS

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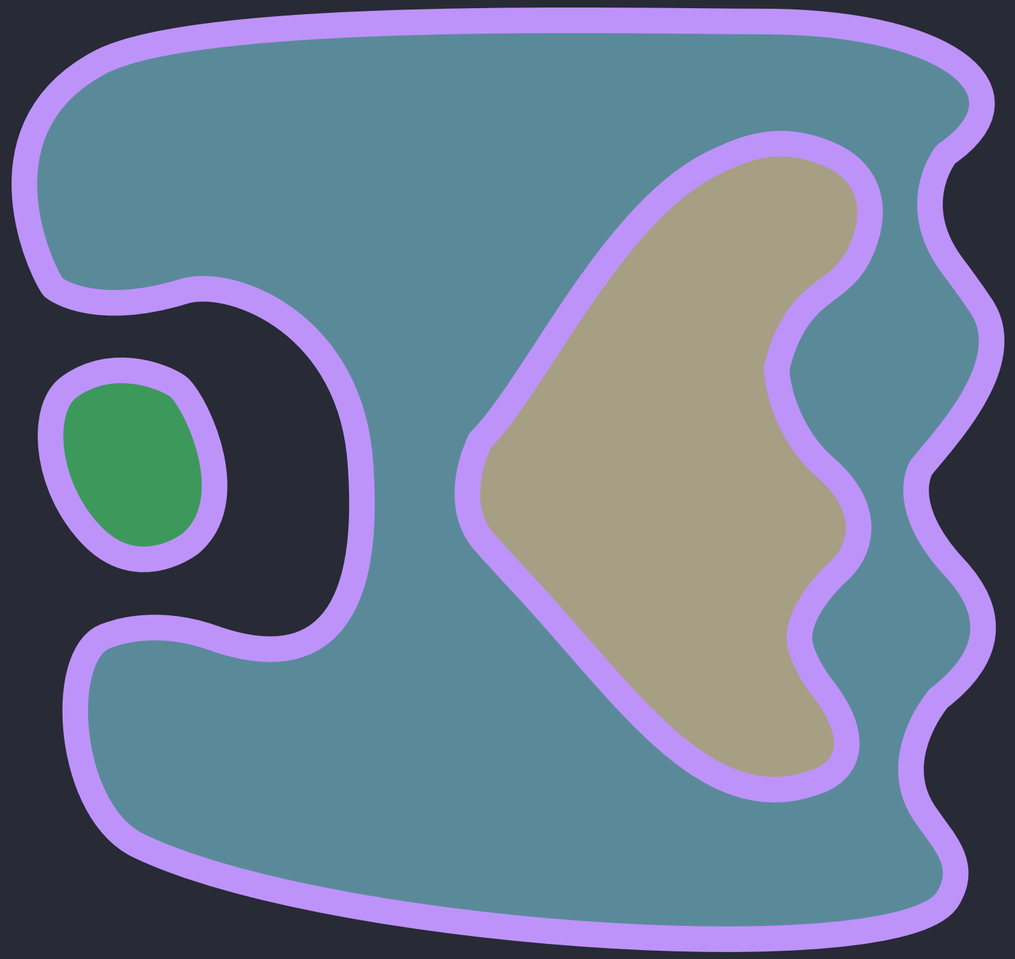
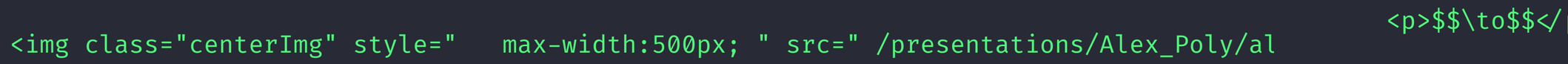


CONSTRUCTION

1. Remove crossings.
2. Connect strands following orientation without creating new crossings.
3. Fill interior of resulting disks.
4. Connect disks with “twists” matching crossing orientation.

SEIFERTVIEW OF 6_2

BANDS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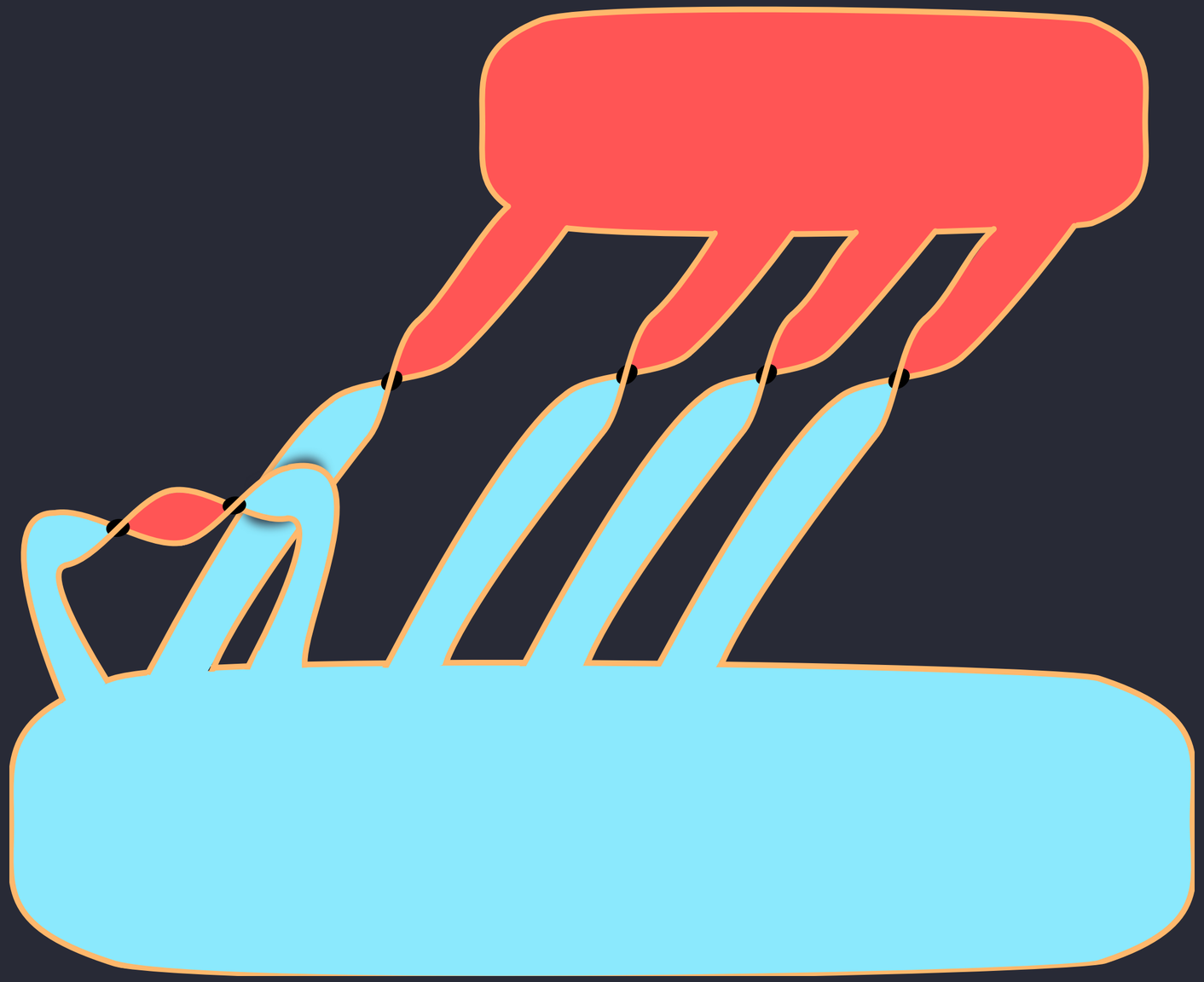

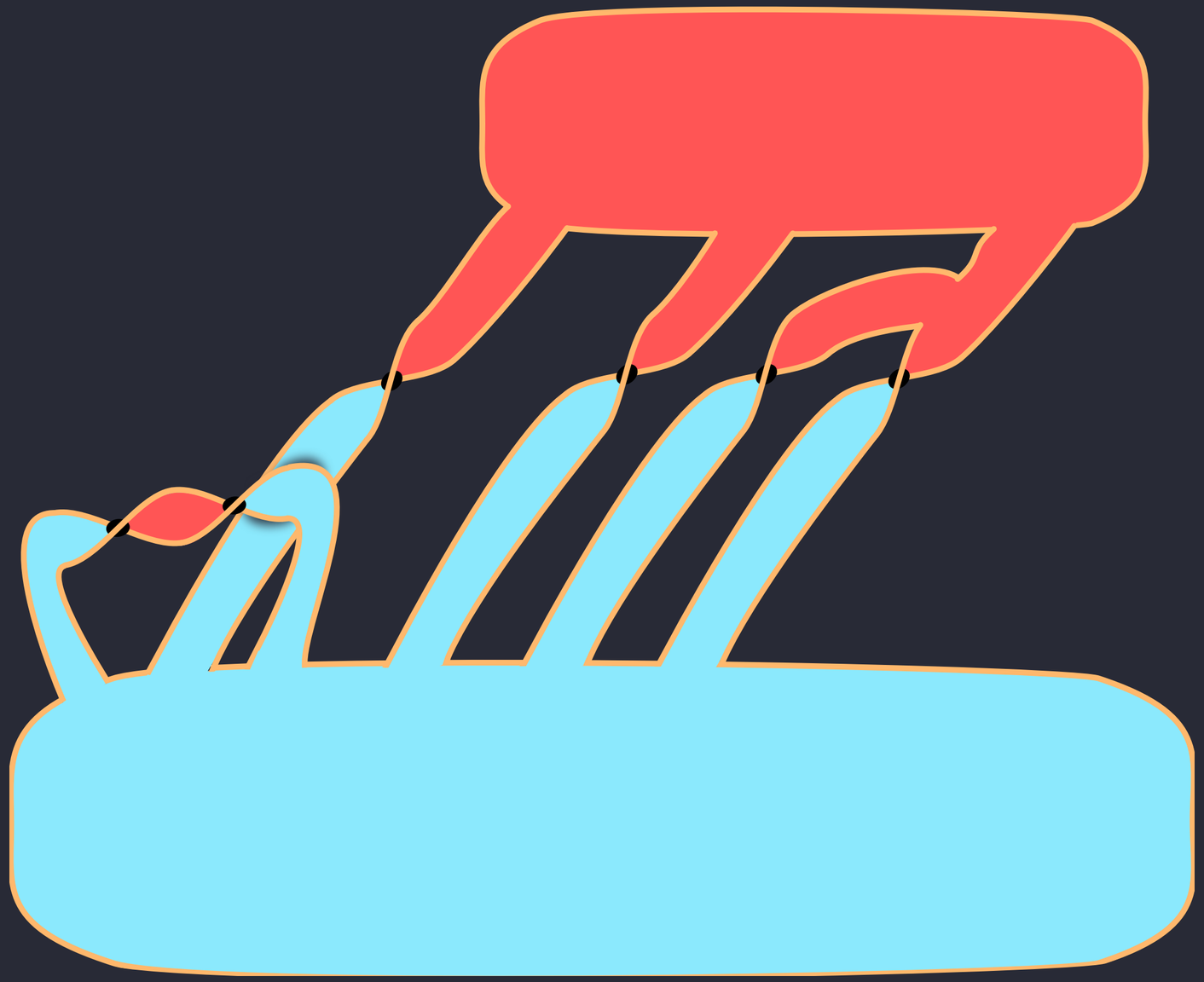

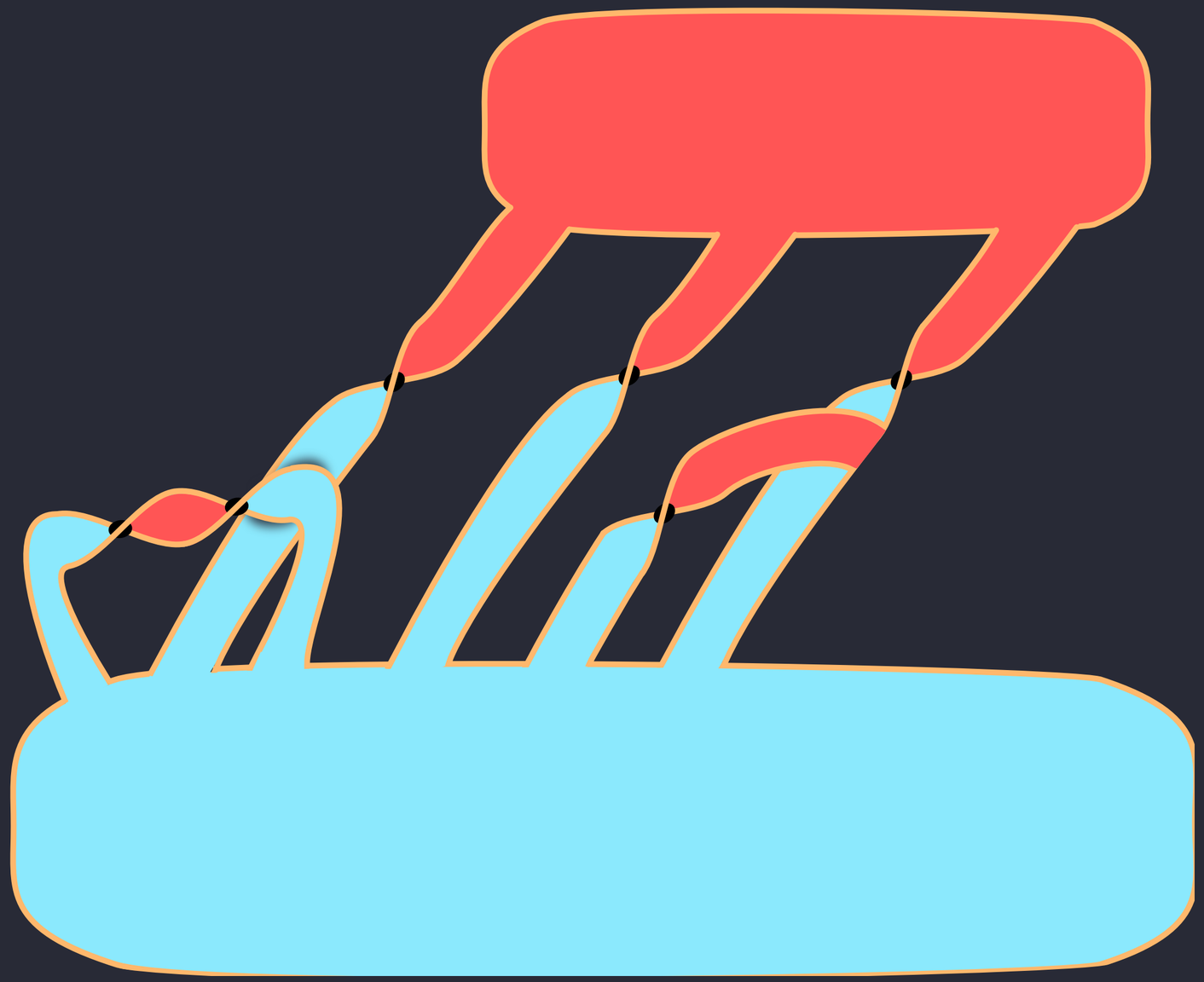

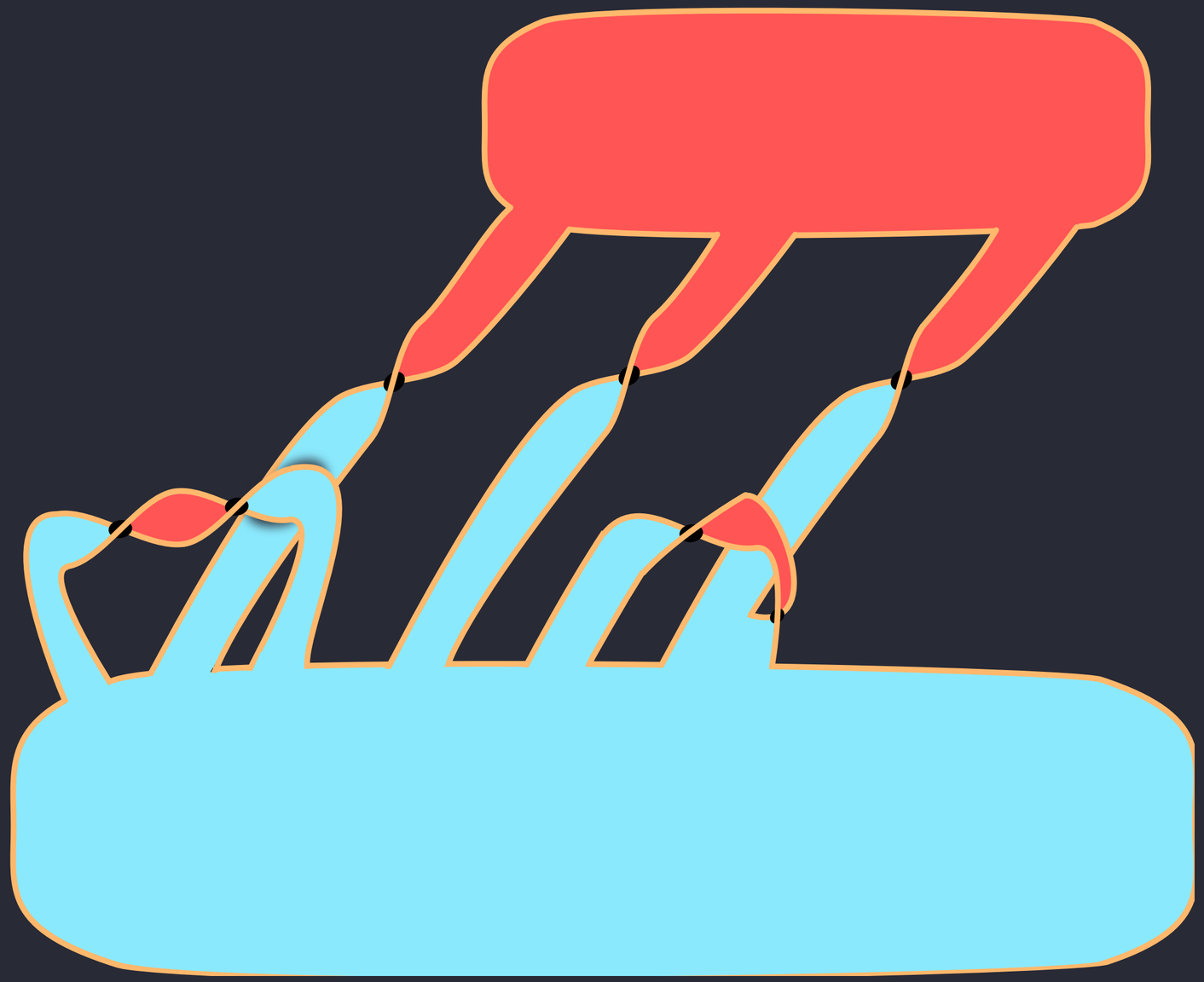

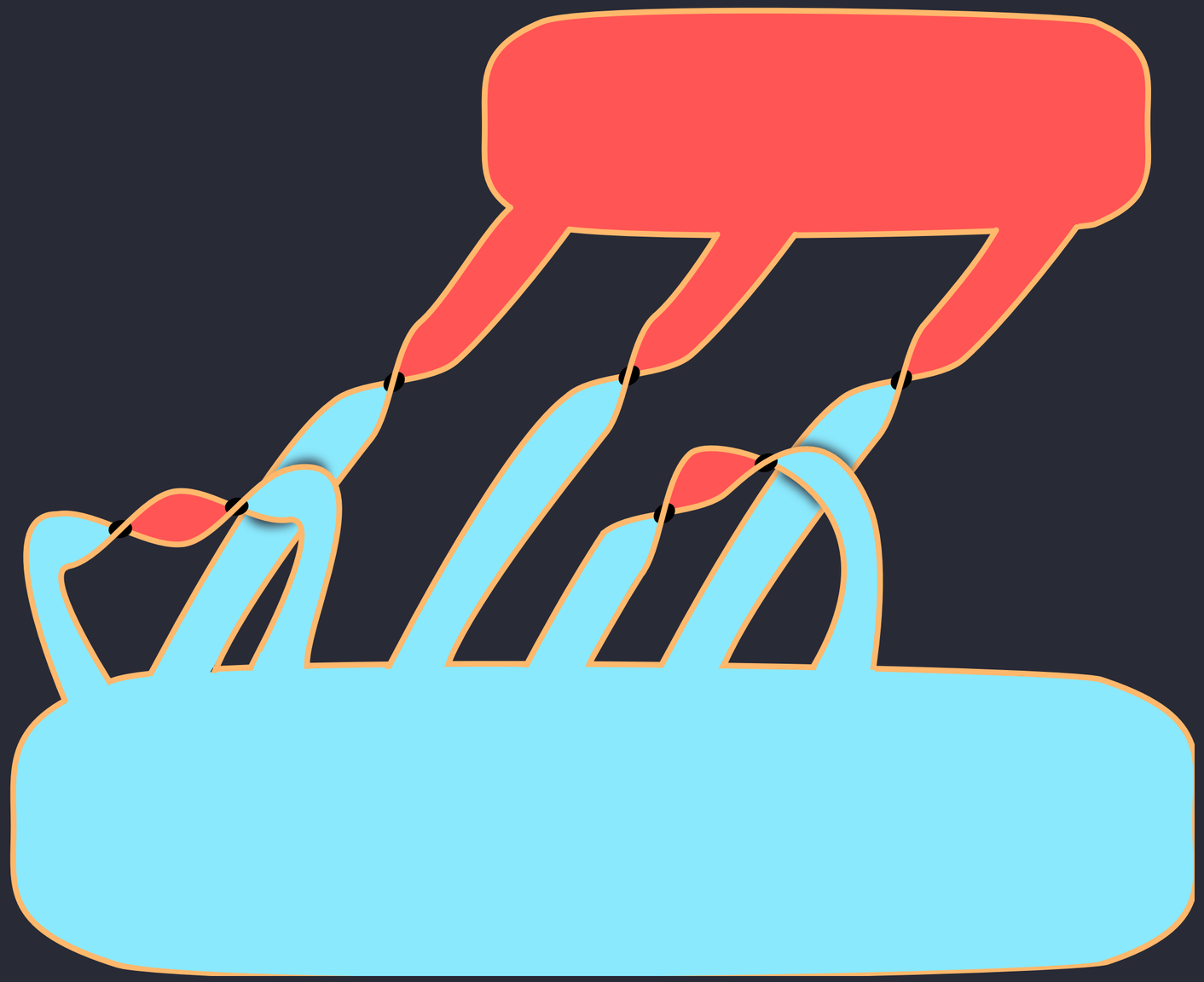

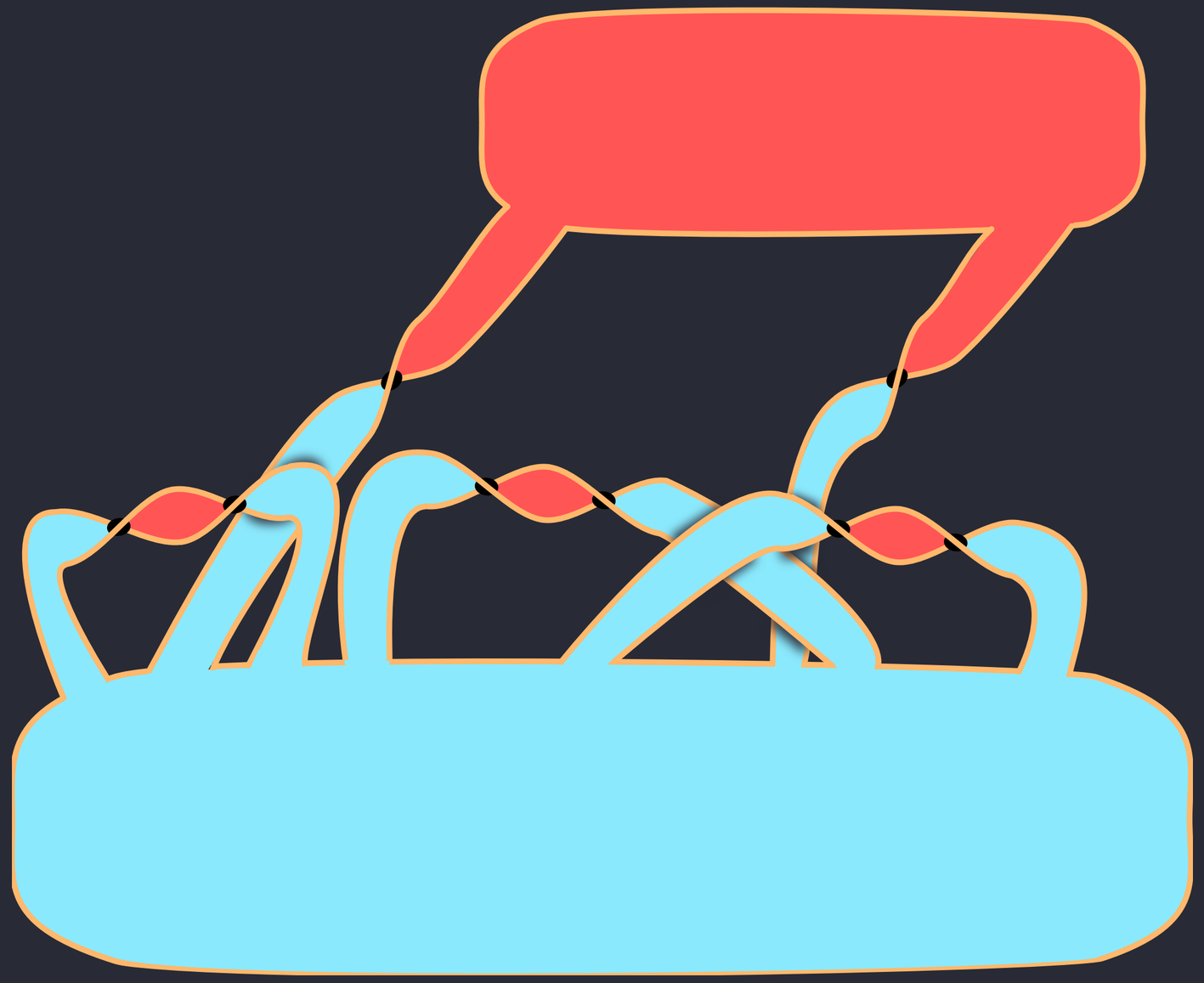

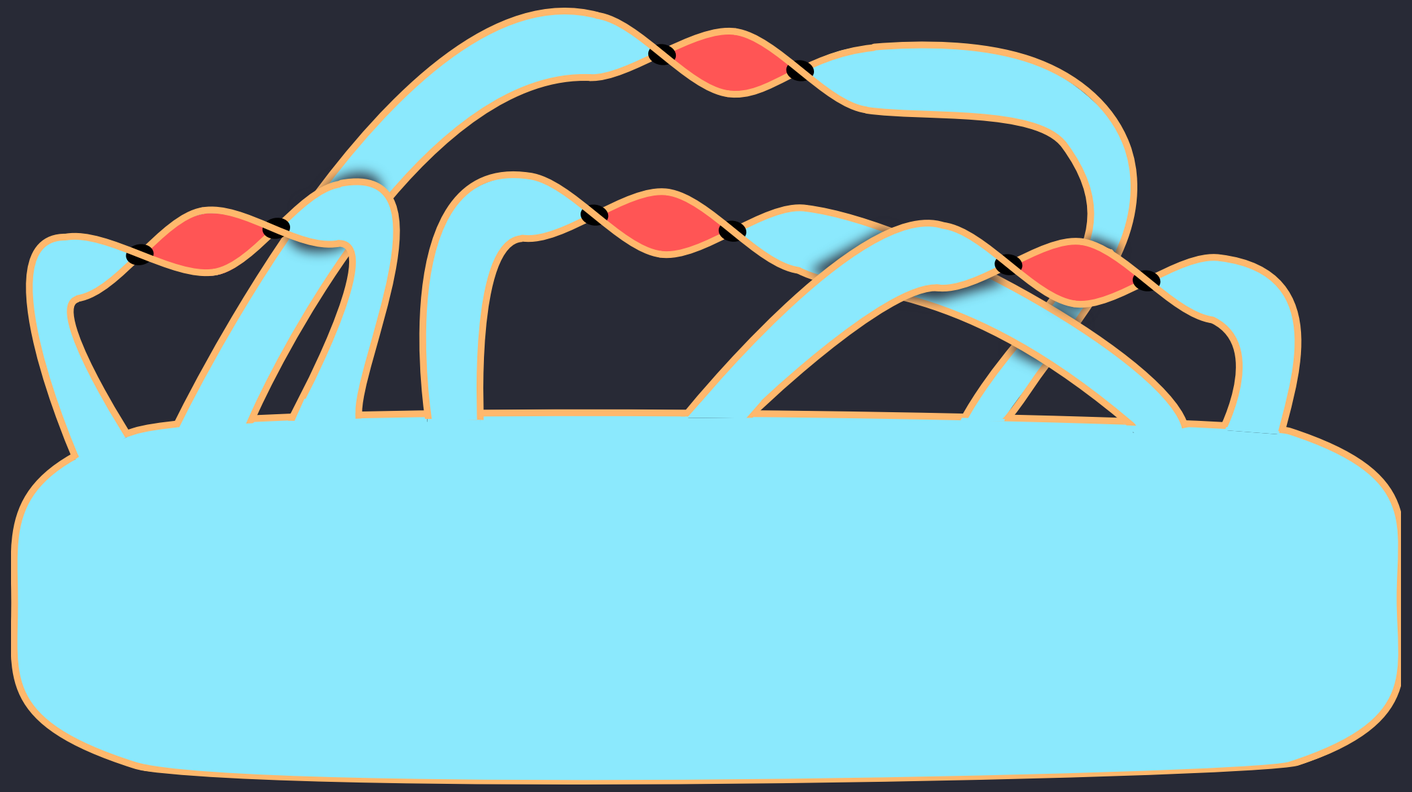

GENUS OF A SEIFERT SURFACE

We have then that as an abstract surface, a Seifert surface for a link is a disc with a number of “handles”

($D^1 \times D^1$) added. That number is its genus.

GENUS OF A LINK

We take the smallest genus of possible Seifert surfaces for a link as **the genus of the link**.

COMPUTING THE GENUS OF A SURFACE

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$$2g = 2 - s - n + c$$

1. g: Genus
2. s: Number of Seifert circles
3. n: Number of components

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SEIFERT MATRIX

LINK CROSSINGS

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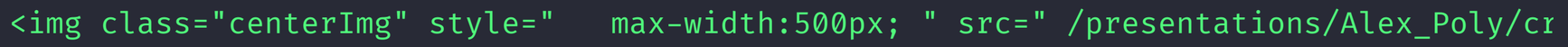
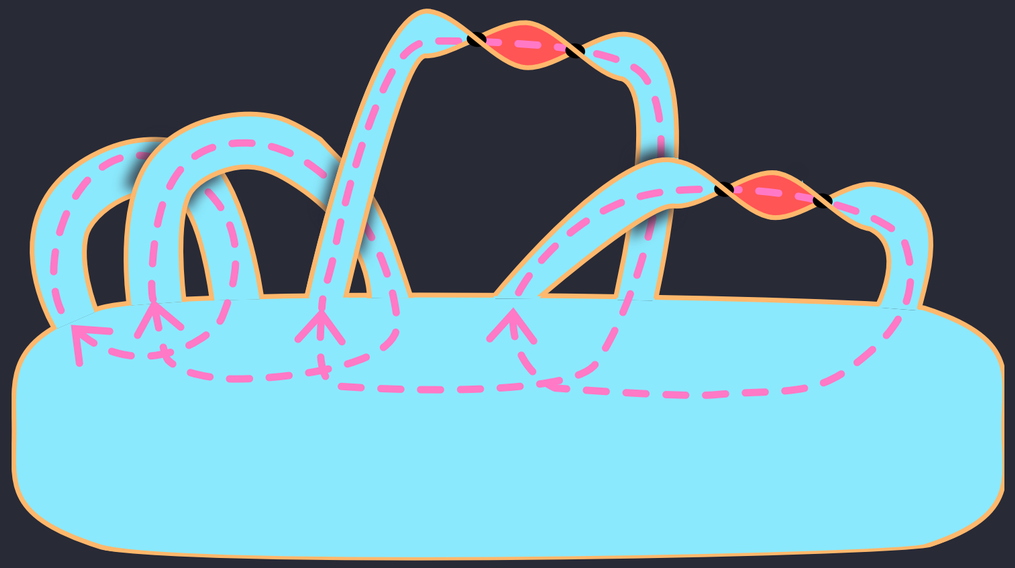


LINKING NUMBER

$$\mathrm{Lk}\left(\mathscr{L}\right) = \#\,\uparrow\!\!\!

LINKS ON A SURFACE

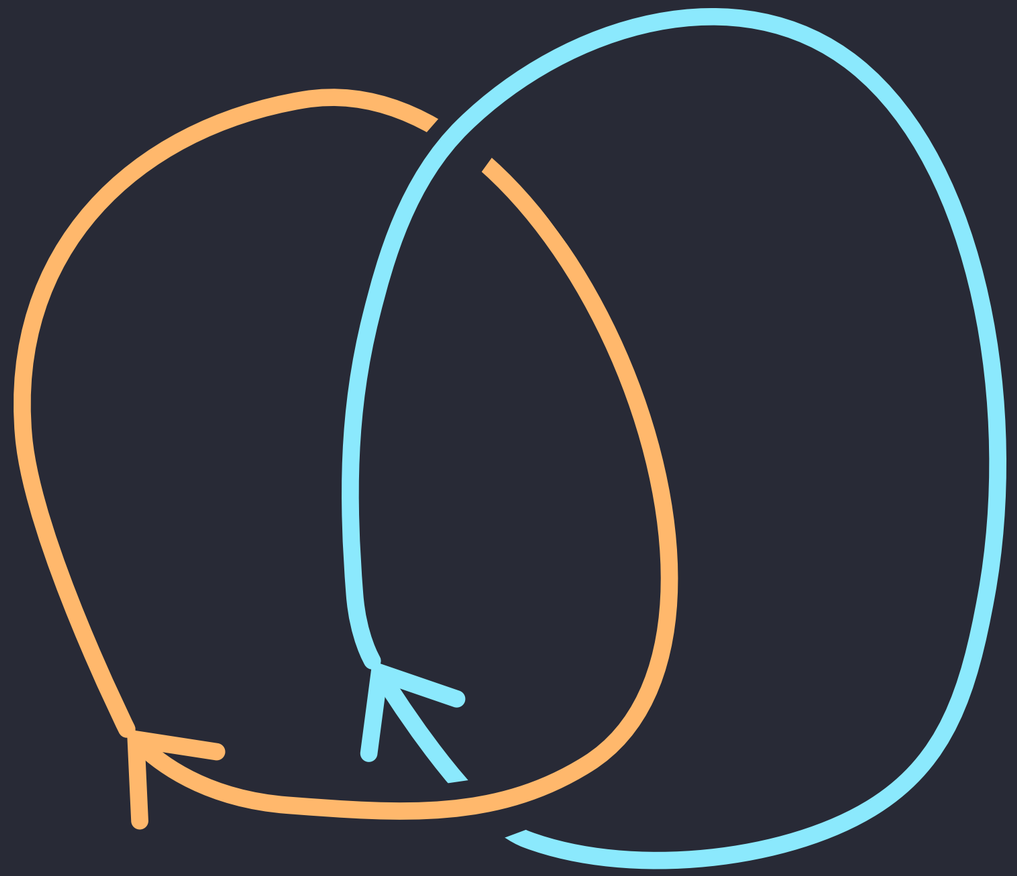
We can put oriented simple closed curves through each of the bands.

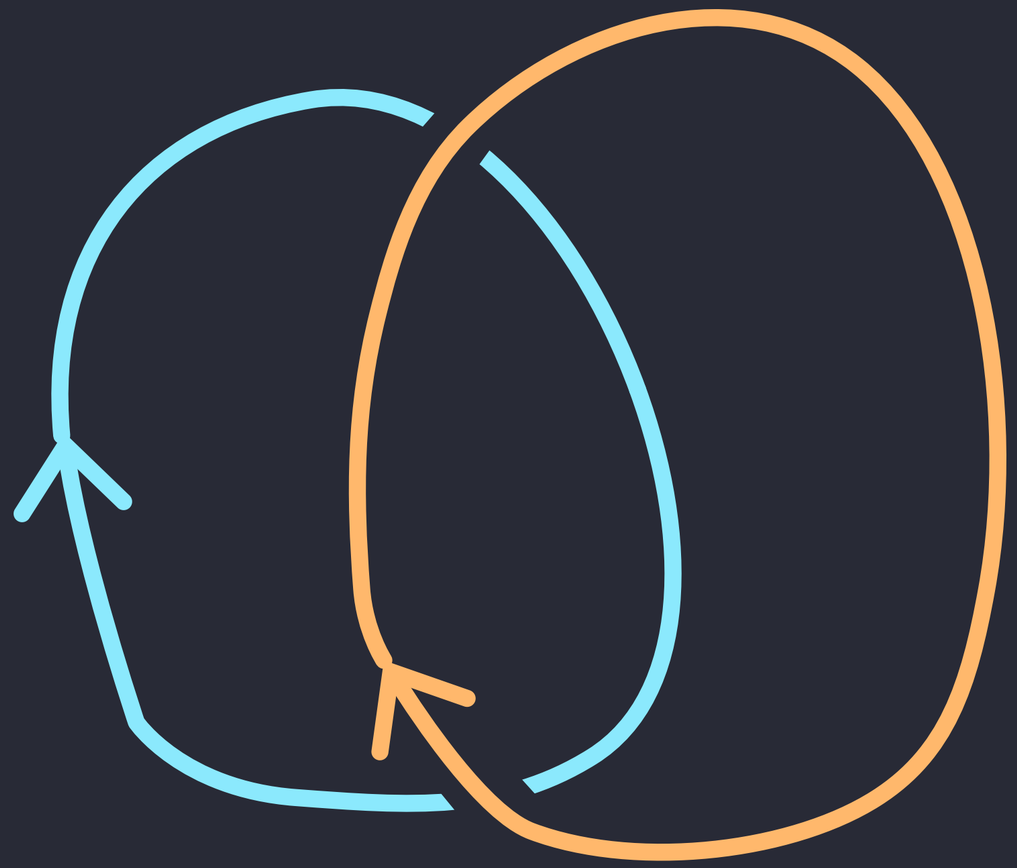


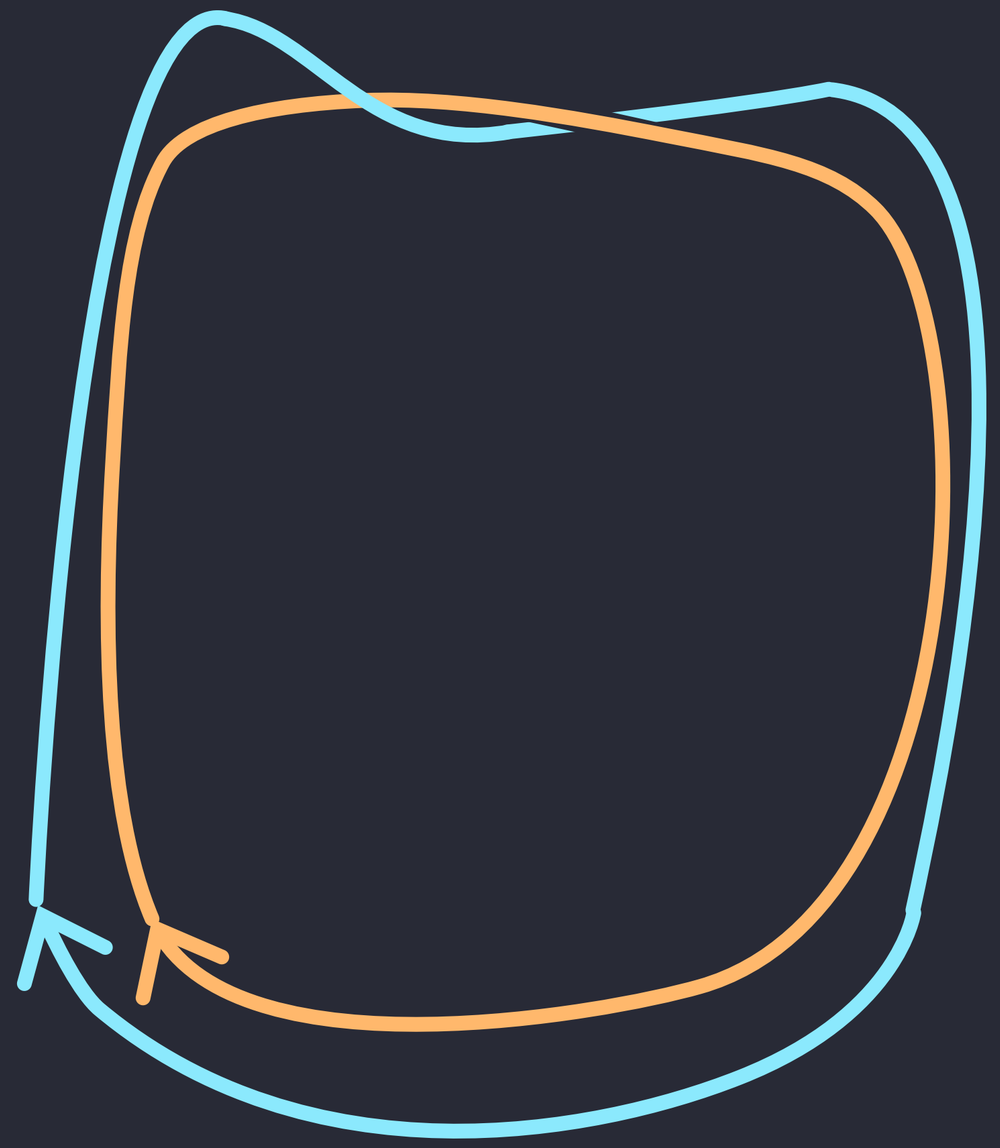
A Seifert surface is oriented, it has a top side and bottom side. We can take a push off of each of the curves in the “up” (blue) direction

We can take an (i, j) pair and compute

$$a_{i,j} = \mathrm{Lk}\left(f_i,\ f_j^+\right)$$







This populates a matrix:

$$\begin{bmatrix} \mathrm{Lk}\left(f_1,\ f_1^+\right) & \mathrm{Lk}\left(f_1,\ f_2^+\right) & \cdots & \mathrm{Lk}\left(f_1,\ f_{2g}^+\right) \\ \mathrm{Lk}\left(f_2,\ f_1^+\right) & \mathrm{Lk}\left(f_2,\ f_2^+\right) & \cdots & \mathrm{Lk}\left(f_2,\ f_{2g}^+\right) \\ \mathrm{Lk}\left(f_3,\ f_1^+\right) & \mathrm{Lk}\left(f_3,\ f_2^+\right) & \cdots & \mathrm{Lk}\left(f_3,\ f_{2g}^+\right) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Lk}\left(f_{2g},\ f_1^+\right) & \mathrm{Lk}\left(f_{2g},\ f_2^+\right) & \cdots & \mathrm{Lk}\left(f_{2g},\ f_{2g}^+\right) \end{bmatrix}$$

EXAMPLE

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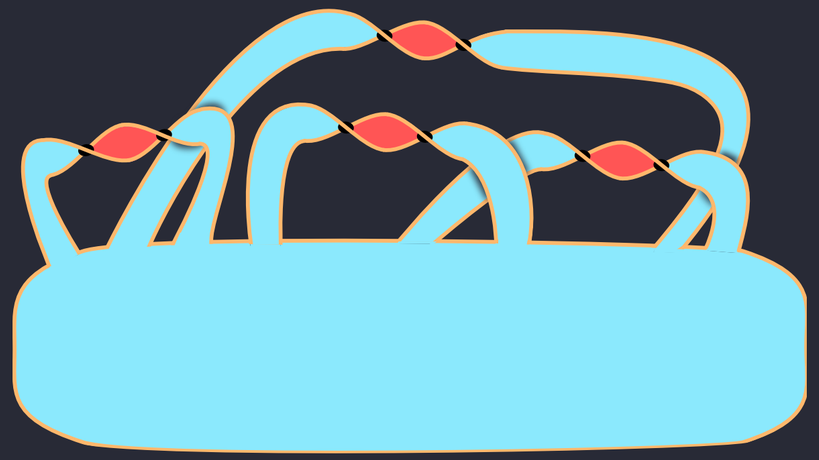
ALEXANDER POLYNOMIAL

For an oriented link $\mathscr{L}$ and it's associated Seifert matrix S we define the *Alexander polynomial* of $\mathscr{L}$ as

$$\Delta_{\mathscr{L}}(t) = \det\left(t^{\frac{1}{2}} S - t^{-\frac{1}{2}} S^T\right)$$

$$\det\left(t^{\frac{1}{2}}\begin{bmatrix}-1 & 1 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & -1 & 0 & 1\end{bmatrix} - t^{\frac{-1}{2}}\begin{bmatrix}-1 & 1 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & -1 & 0 & 1\end{bmatrix}^T\right) = -t^4 + 3t^3 - 3t^2 + 3t - 1$$



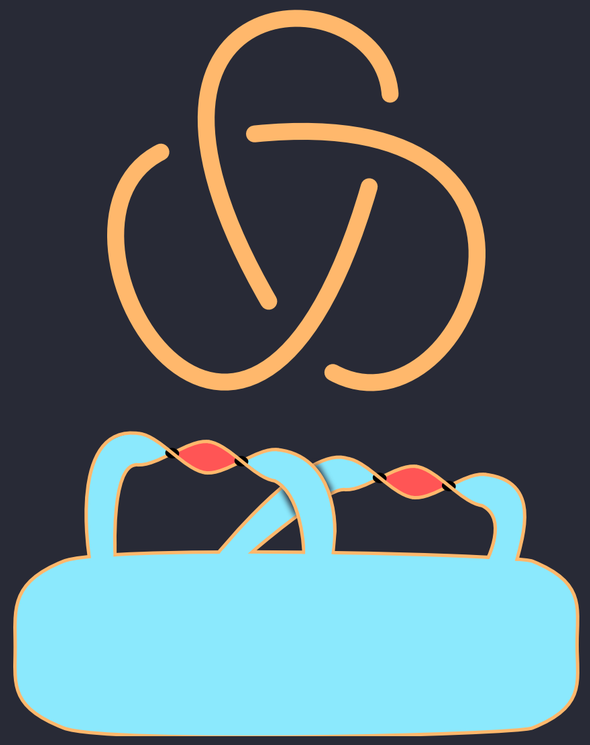
$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

$$-t^4 + 3t^3 - 3t^2 + 3t - 1$$

INVARIANT

$\Delta_{\mathscr{L}}(t)$ is unique up to **stabilization**, a method for adding bands to the surface. Results in $\Delta_{\mathscr{L}}(t)$ being unique up to a $\pm t^k$.

LIMITATIONS: EXAMPLE



$$1 - t + t^2$$

$$1 - t + t^2$$

BOUND ON GENUS

$$\deg\left(\Delta_{\mathscr{L}}(t)\right) \leq 2g$$

SOURCES

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5. van Wijk, J. J., & Cohen, A. M. (2006). Visualization of Seifert surfaces. In IEEE Transactions on Visualization and Computer Graphics (Vol. 12, Issue 4, pp. 485-496). Institute of Electrical and Electronics Engineers (IEEE). https://doi.org/10.1109/tvcg.2006.83
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